

Virtual Metals Trading Desk — one-page summary

A simulated physical aluminium-scrap import desk into India, built to be defended in a physical-trading interview. It buys Zorba 95/5, Taint/Tabor and Tense on Jebel Ali → Nhava Sheva and US East Coast → Mundra, sells to Indian smelters and hedges on MCX and in USD/INR forwards, through the real 1-Mar-2022 to 31-Aug-2022 market, including the 7-Mar LME record close and the crash after it.

ACADEMIC SIMULATION — not actual trades. Counterparties, vessels, banks, tickets and events are fictional (SIM). Market data is real where a public source exists, otherwise flagged PROXY or ASSUMPTION. Every number is filled from the published tables by code, with its source file named.

The desk in numbers

Tickets	Tonnes	Containers	Grades · lanes	Counterparties	Traded	Ex-ante trade rule
9	14,350 MT	656	3 · 2	3 suppliers, 3 buyers	8-Mar to 3-Aug-2022	9 of 9 tickets pass

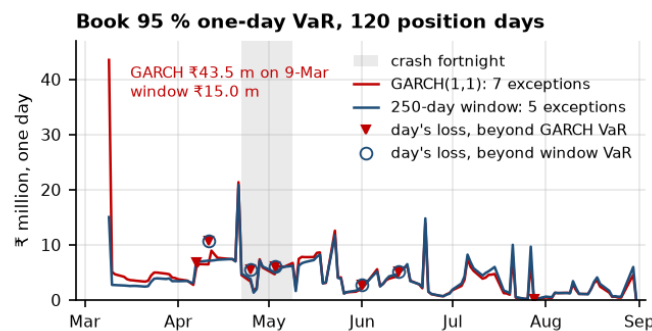
Rule declared before any parity result (CONTRACTS §5a): parity window open on the base screen, on the point-in-time grade mix and with conversion cost a step higher. 82 of 162 weekly cases passed (111 on the base screen alone) (*parity_weekly.csv*).

Headline results

Result	Value and source
Book P&L to the horizon (31-Oct-2022)	₹192.1 m (₹13,384/MT); ₹195.9 m at the window end (31-Aug-2022) (<i>attribution_daily.csv</i>)
...read only with its band	—₹105.4 m to +₹334.3 m across the registered domestic anchor_premium_inr_t grid (ASSUMPTION, verification PENDING); break-even —38,702 ₹/MT inside the grid, so the headline is not sign-robust (<i>pnl_sensitivity_sign_robustness.csv</i>)
Where it came from	deal margin at contract dates +₹165.2 m, grade spread +₹103.3 m, carry, roll and cross-terms —₹56.6 m, LME flat price —₹30.0 m; the other 4 buckets +₹10.0 m; residual ₹0. Deal margin (86 % of the total) comes from the desk's own sale-pricing rule on that anchor premium (<i>attribution_daily.csv</i>)
Best trade (post-mortem metric)	T02, Tense, Jebel Ali → Nhava Sheva: +₹50.55 m (₹42,122/MT). Ranked on worst P&L across 24 registered re-pricing cases, its floor is +₹10.65 m: the only ticket positive in every case. T01 makes more in rupees (+₹64.28 m) but falls to —₹28.93 m in its worst case (<i>pnl_sensitivity_pricing.csv</i>)
Worst trade	T08, Tense, US East Coast → Mundra: —₹11.73 m (—₹6,980/MT); hit its hard stop on 1-Aug (<i>attribution_daily.csv</i>)



ACADEMIC SIMULATION — not actual trades
attribution_daily.csv; band: *pnl_sensitivity_sign_robustness.csv*



ACADEMIC SIMULATION — not actual trades
var_daily.csv; LME DIRECT, USD/INR and MCX PROXY; book SIM

Risk findings, as they came out

Model	Finding and source
Daily 95 % VaR: GARCH(1,1) vs 250-day history	Mixed, not the win the spec expected. Into March GARCH moved hardest (LME vol forecast for 8-Mar 2.89 % against 1.67 %; book VaR ₹43.5 m on 9-Mar against ₹15.0 m). But on the declared 90th-percentile alert rule the 250-day window crossed first (9-Feb, 3 trading days before GARCH), and into the 22-Apr crash fortnight GARCH (1.85 %) sat below both windows (1.96 % at 250 days, 2.84 % at 60). Mean VaR, 120 days: ₹4.27 m against ₹3.98 m (<i>var_summary.csv</i> , <i>var_lead_lag.csv</i>)
Kupiec backtest	Book: 7 GARCH and 5 window exceptions against 6 expected. Neither is rejected, but the test accepts 2 to 11 here, so it cannot separate them. On a fixed LME long, 2019–2022, neither is rejected either; taking 2021 alone, the 250-day window is rejected and GARCH is not (<i>kupiec.csv</i>)
Monte Carlo, 10,000 paths, 99 % 10-day VaR	On three dates fixed by rule: ₹76.3 m on 8-Mar (no MCX hedge yet), ₹23.0 m on 21-Apr, ₹1.00 m on 27-Jul. An MCX basis factor lifts the last two to ₹32.8 m and ₹14.6 m: on the near-flat 27-Jul book, basis is most of the risk. Beyond every path: BIS-QCO hold + demurrage (HYPOTHETICAL policy) on 21-Apr (—₹45.6 m) and 27-Jul (—₹18.3 m); BUY_RJK_01 (SIM) defaults on 27-Jul (—₹28.3 m) (<i>mc_summary.csv</i> , <i>mc_stress_scenarios.csv</i>)
Credit scoring (logistic)	Riskiest first: BUY_RJK_01 PD 41.8 %, band D; BUY_JNPT_01 8.2 % (band C); BUY_MUN_01 5.1 % (band C). BUY_RJK_01's contracted exposure reached 2.57× its line, which the model cuts from ₹120 m to ₹60 m. Illustrative: fitted on 1,200 synthetic buyer-quarters; profiles written by the book's author (<i>credit_scores.csv</i>)
Liquidity and margin	The book did not fit its bank lines. Funding need peaked at ₹1,179.0 m on 20-Jul against a ₹1,000 m line sized on the plan's average balance: 19 days over (none on a line one ₹25 crore step higher). LCs peaked at ₹1,414.8 m against ₹1,000 m (17 days over, face plus tolerance). Peak MCX initial margin ₹95.8 m (21-Apr) (<i>margin_liquidity_summary.csv</i>)
Sentiment overlay (VADER)	649 real headlines over 28 weeks: 0 of 12 lead tests significant, so no evidence that headline tone led LME. The March spike could not be tested, and 28 weeks can only rule out a strong lead (<i>sentiment_summary.csv</i>)

What is real, a proxy or an assumption

Count	DIRECT	PROXY	ASSUMPTION	Total
Parameters (register)	51	16	165	232
Market-panel columns	7	14	7	28

Parameter verification: 47 VERIFIED, 24 PARTIAL, 46 PENDING, 115 N/A (desk policy or model design).

- Biggest proxy: MCX** at duty-paid import parity, one-for-one with LME × USD/INR, so hedges look perfect. A third-party mirror gives weekly beta 0.75; at that beta mean GARCH VaR is ₹8.5 m, not ₹4.27 m.
- Reconstructed: freight and grade factors.** Freight levels are hindsight-calibrated; the grade path uses later data. On the point-in-time grade mix the book makes ₹97.3 m and grade spread turns —₹36.8 m.

Where to look

- README.md (method, re-run steps, caveats) and docs/INDEX.md (every spec row mapped to its files).
- In outputs/reports/: 6 weekly desk notes, the 2-page post-mortem on T02, the one-page risk policy memo and the 17-question interview pack.
- Metals_Desk_Master.xlsx: formula-driven parity, book and P&L attribution; reconciliation **VERIFIED**: 132,447 formula cells recalculated outside Excel, 0 failed checks.

Built with

Python (pandas, arch, SciPy, scikit-learn, matplotlib, reportlab, VADER); formula-driven Excel (openpyxl, recalculated outside Excel with formulas). Seeded, staged through files: DESK_OFFLINE=1 python run_all.py rebuilds everything offline from cached raw data.